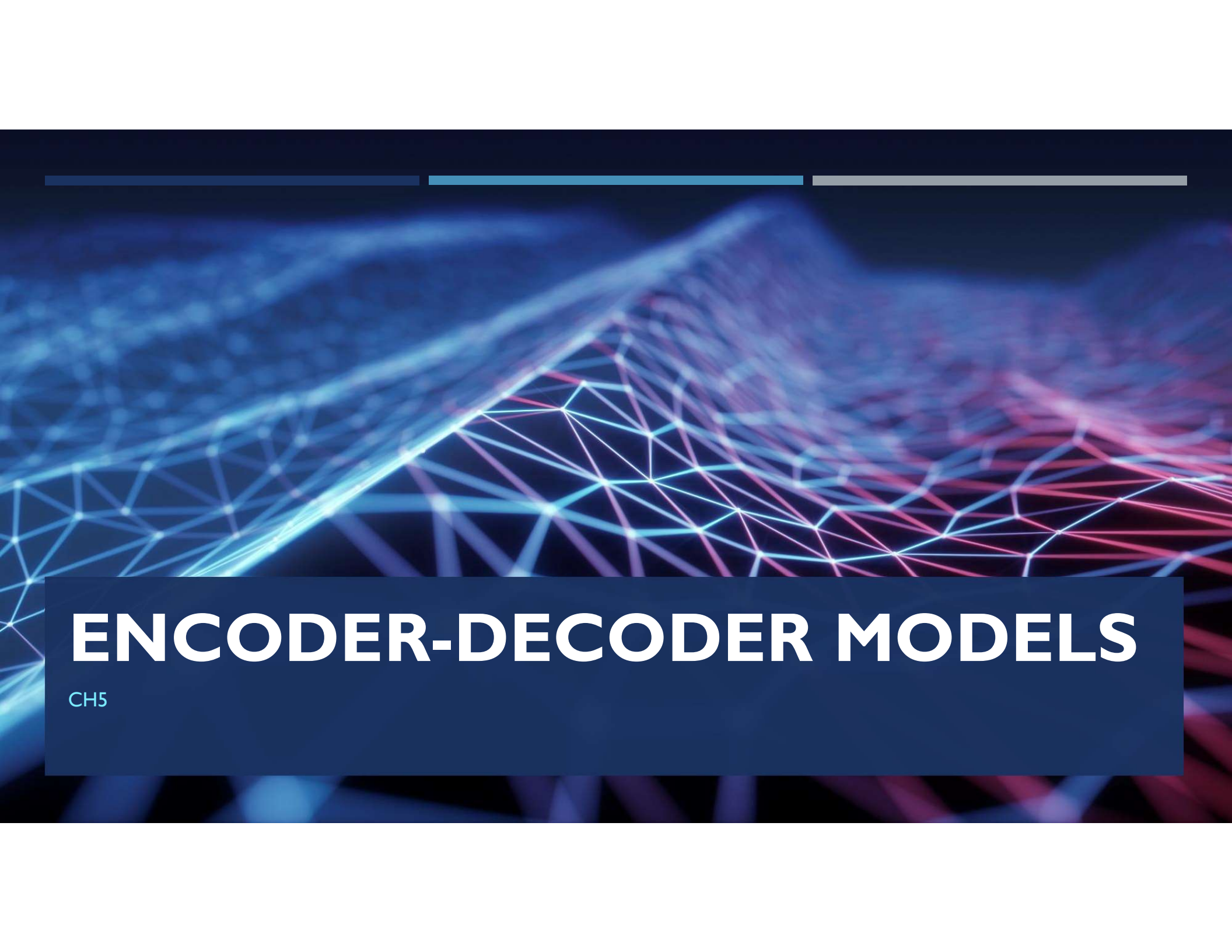




ENCODER-DECODER MODELS

CH5

WHICH FACE IS REAL



A



B



C

SUPERVISED LEARNING VS UNSUPERVISED LEARNING

Supervised Learning

Data: (x, y)
 x is data, y is label

Goal: Learn function to map
 $x \rightarrow y$

Examples: Classification, regression, object detection, semantic segmentation, etc.

Unsupervised Learning

Data: x
 x is data, no labels!

Goal: Learn some *hidden* or *underlying structure* of the data

Examples: Clustering, feature or dimensionality reduction, etc.

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Goal: Take as input training samples from some distribution and learn a model that represents that distribution

Density Estimation



Sample Generation



Input samples

Generated samples

Training data $\sim P_{data}(x)$

Generated $\sim P_{model}(x)$

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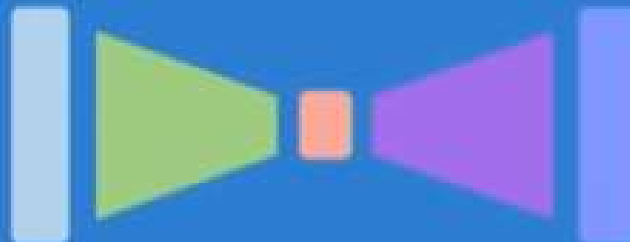
Generated samples

Generated $\sim P_{model}(x)$

How can we learn $P_{model}(x)$ similar to $P_{data}(x)$?

Latent variable models

Autoencoders and Variational
Autoencoders (VAEs)

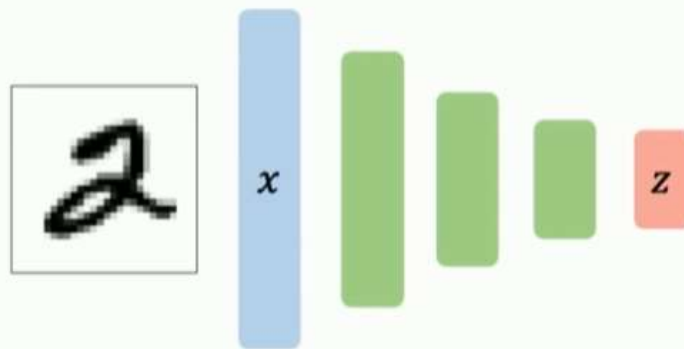


AUTOENCODERS



AUTOENCODERS: BACKGROUND

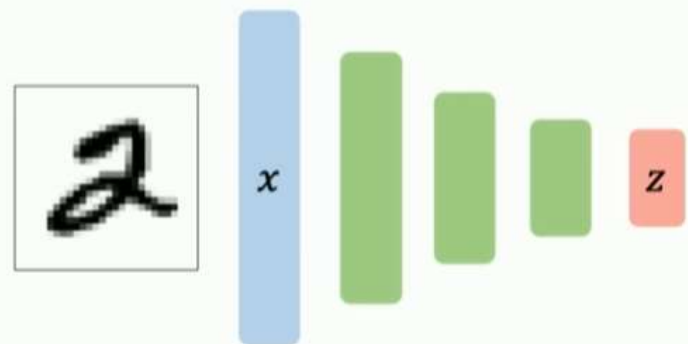
Unsupervised approach for learning a **lower-dimensional** feature representation from unlabeled training data



"Encoder" learns mapping from the data, x , to a low-dimensional latent space, z

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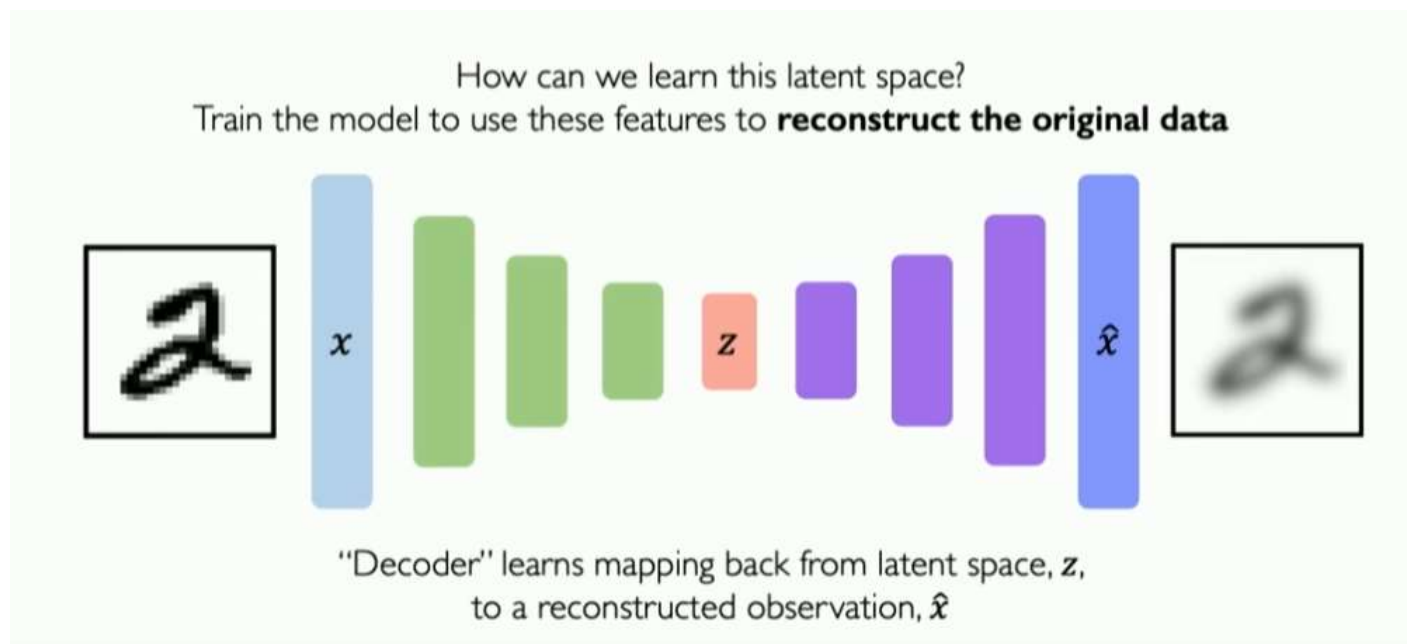
Unsupervised approach for learning a **lower-dimensional** feature representation from unlabeled training data



Why do we care about a low-dimensional z ? 🤔

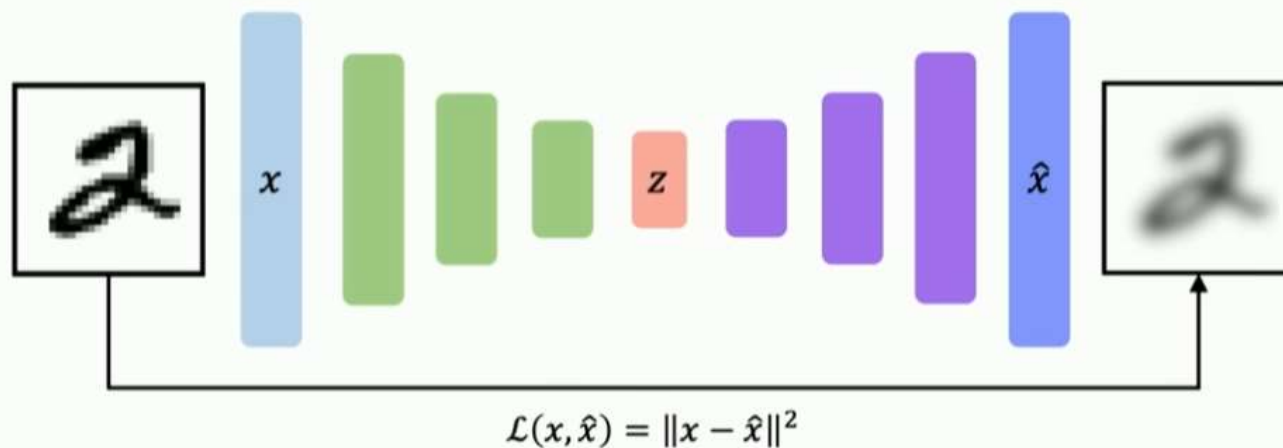
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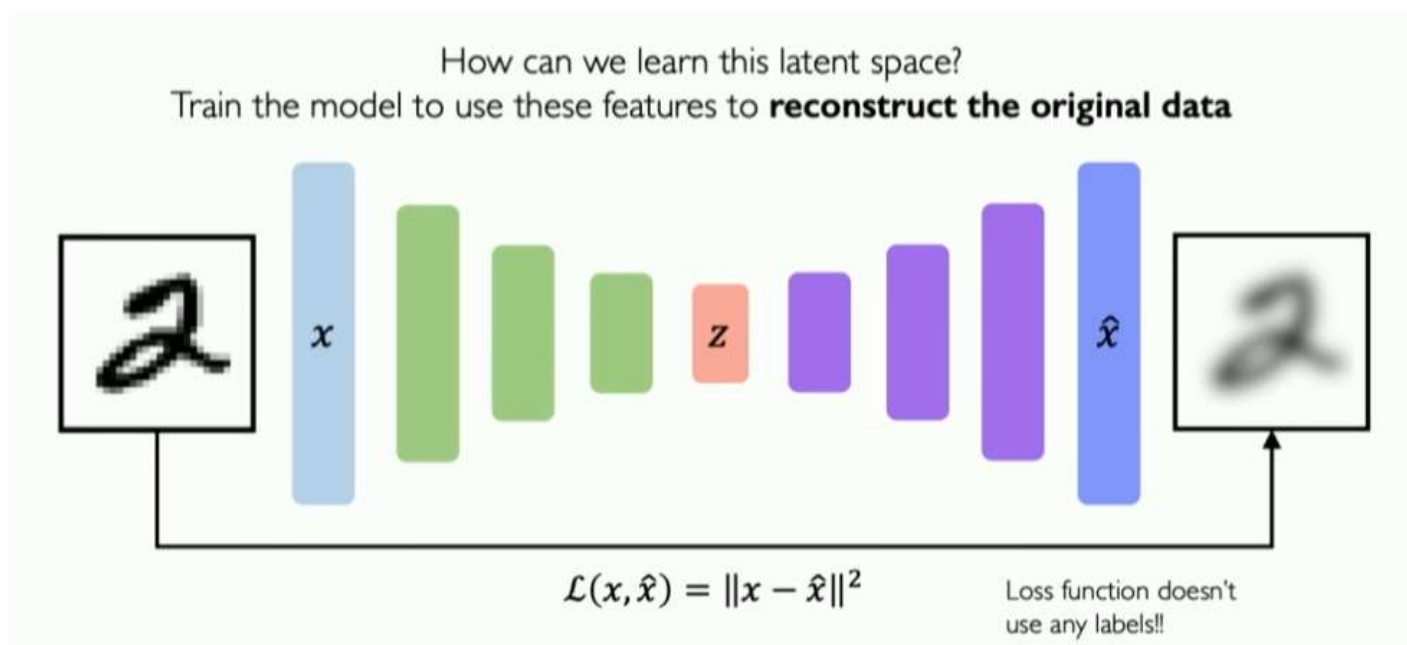


AUTOENCODERS: BACKGROUND

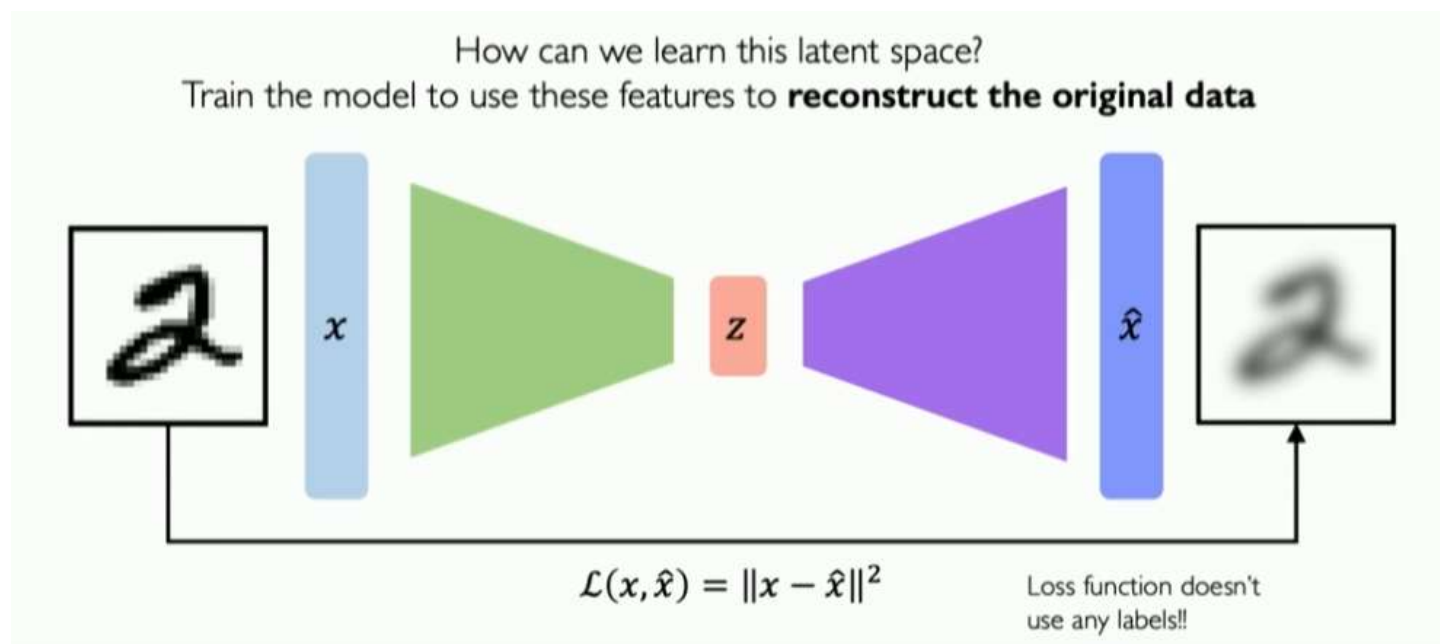
How can we learn this latent space?
Train the model to use these features to **reconstruct the original data**



AUTOENCODERS: BACKGROUND



AUTOENCODERS: BACKGROUND



Dimensionality of latent space → reconstruction quality

Autoencoding is a form of compression!
Smaller latent space will force a larger training bottleneck

2D latent space



5D latent space



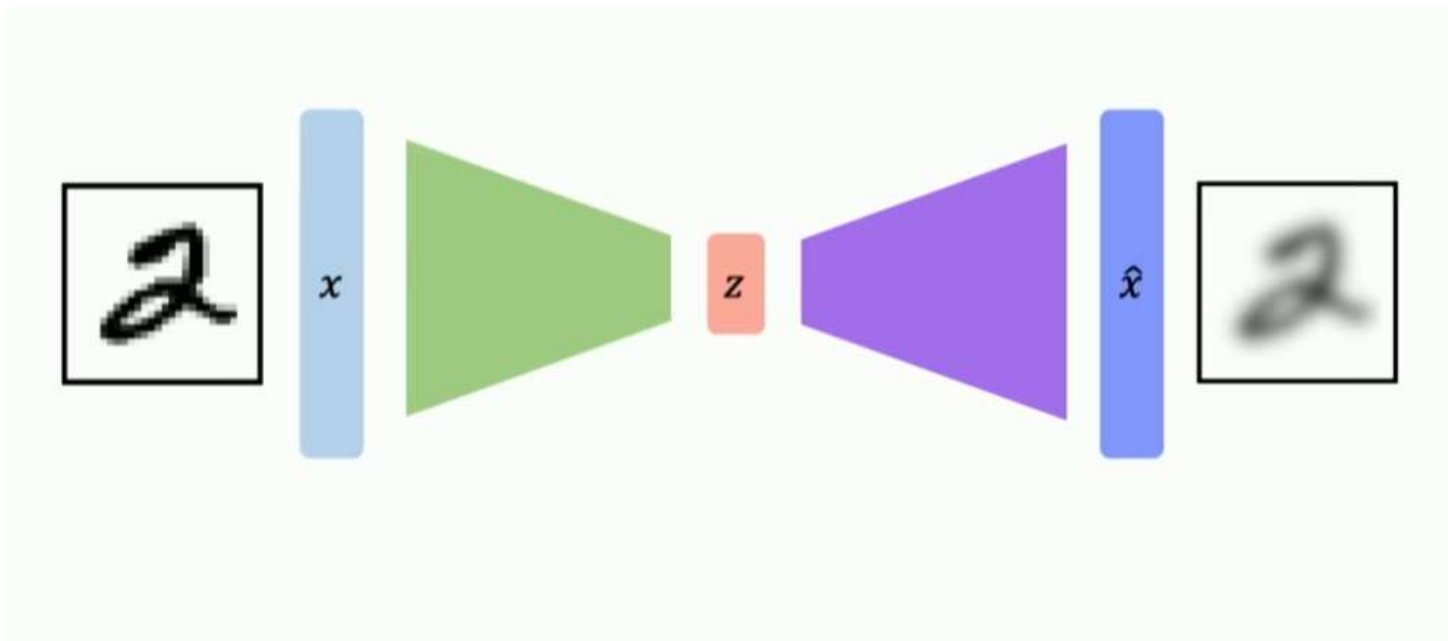
Ground Truth



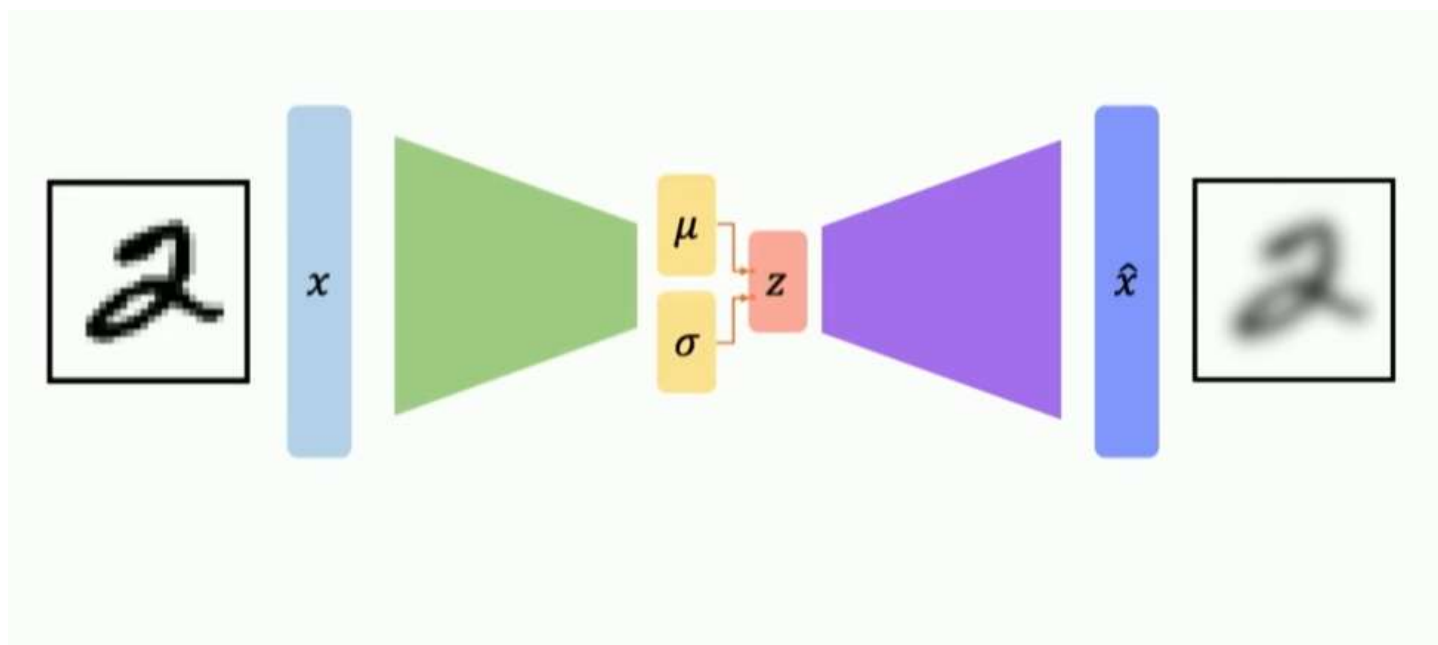
VARIATIONAL AUTOENCODERS (VAE)



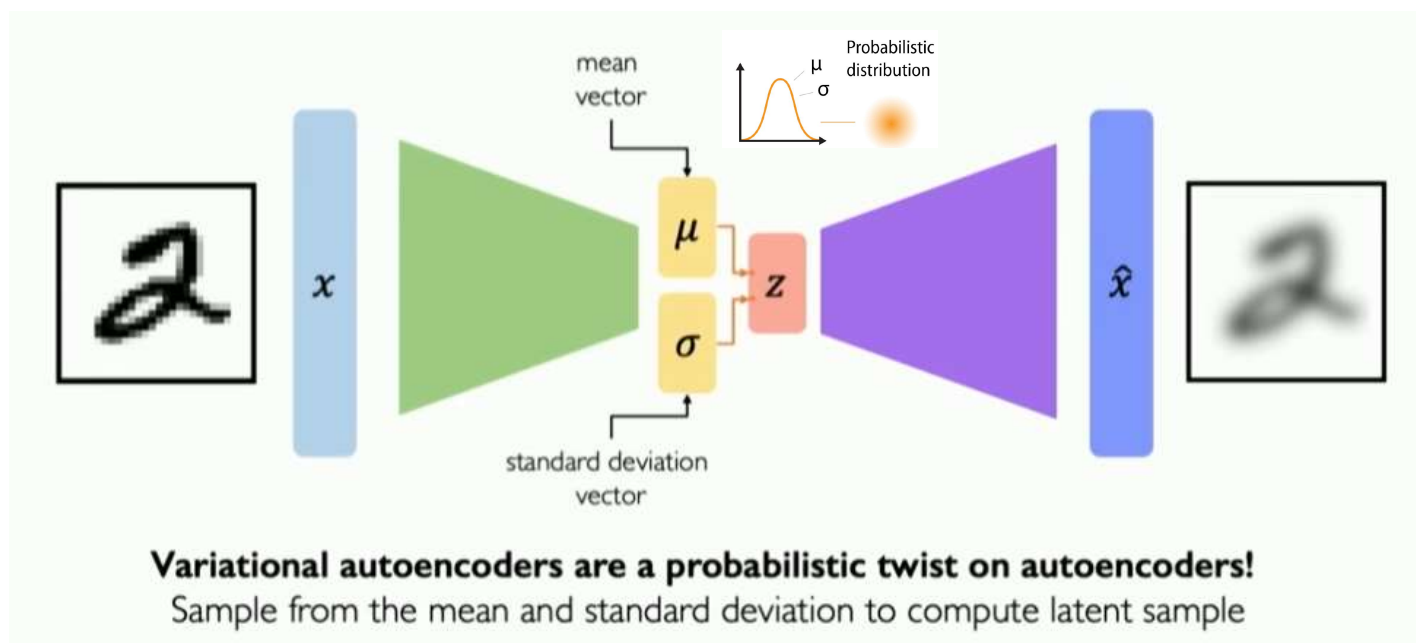
TRADITIONAL AUTOENCODERS



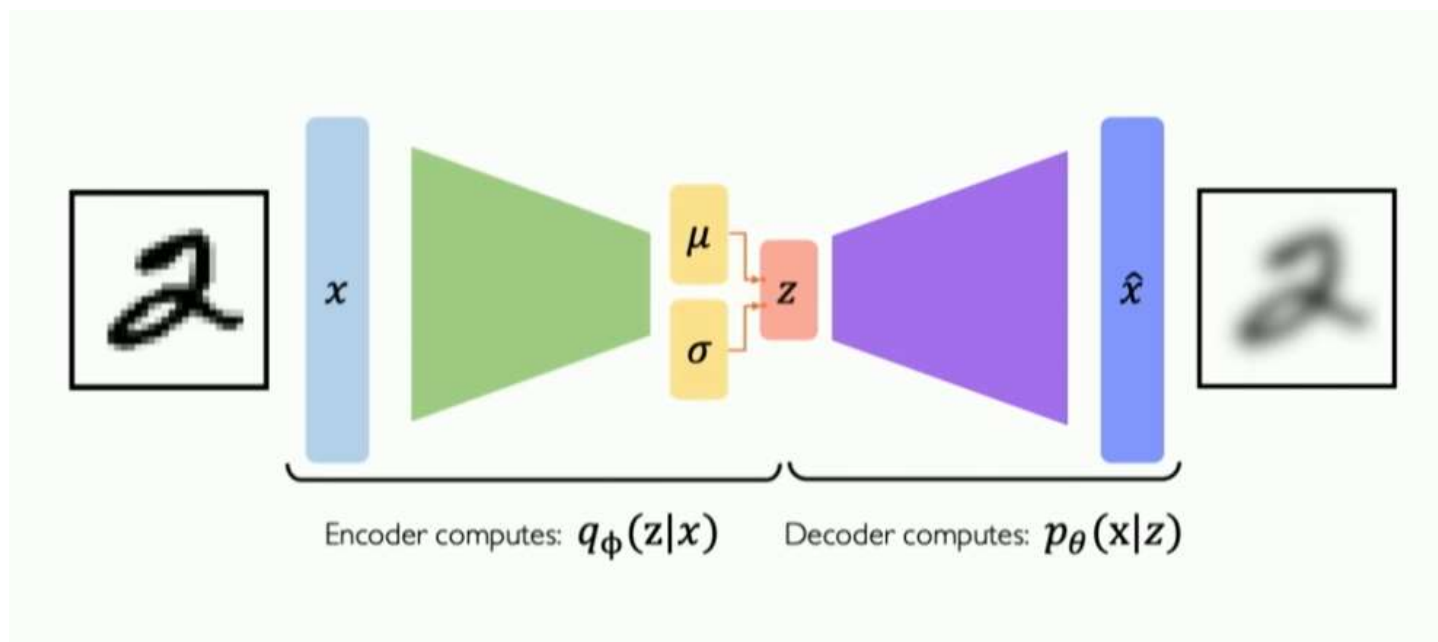
VAE: KEY DIFFERENCE WITH TRADITIONAL AUTOENCODER



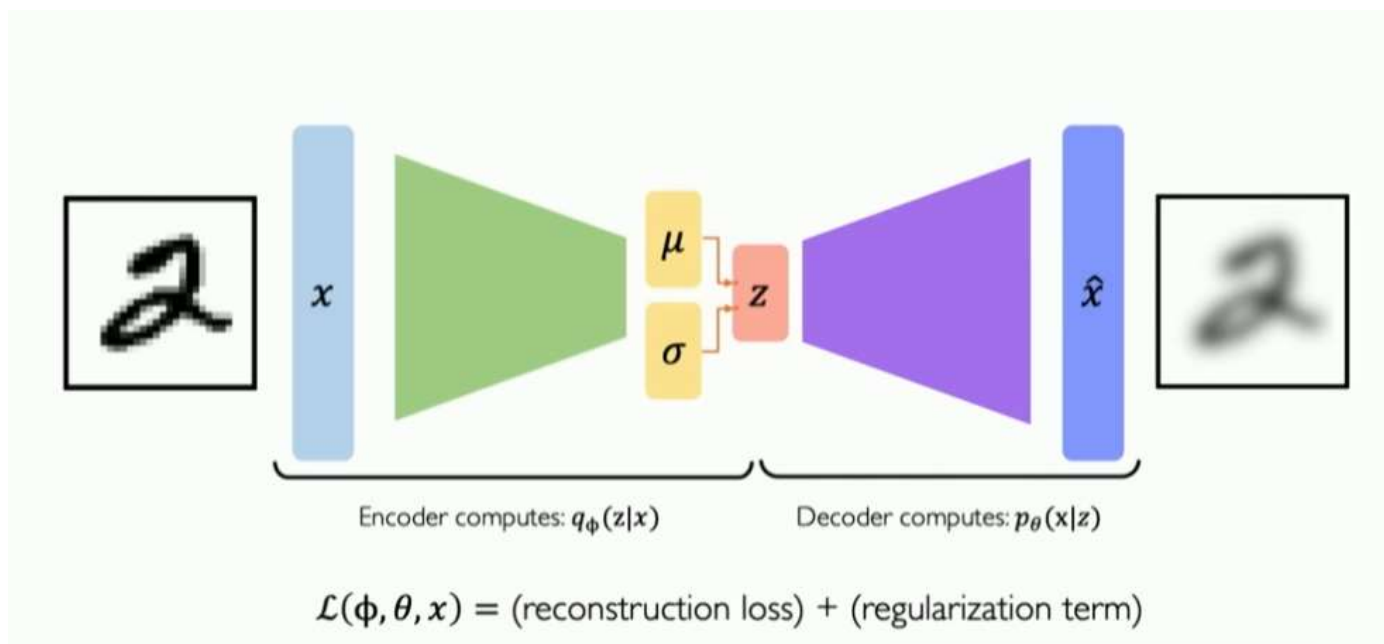
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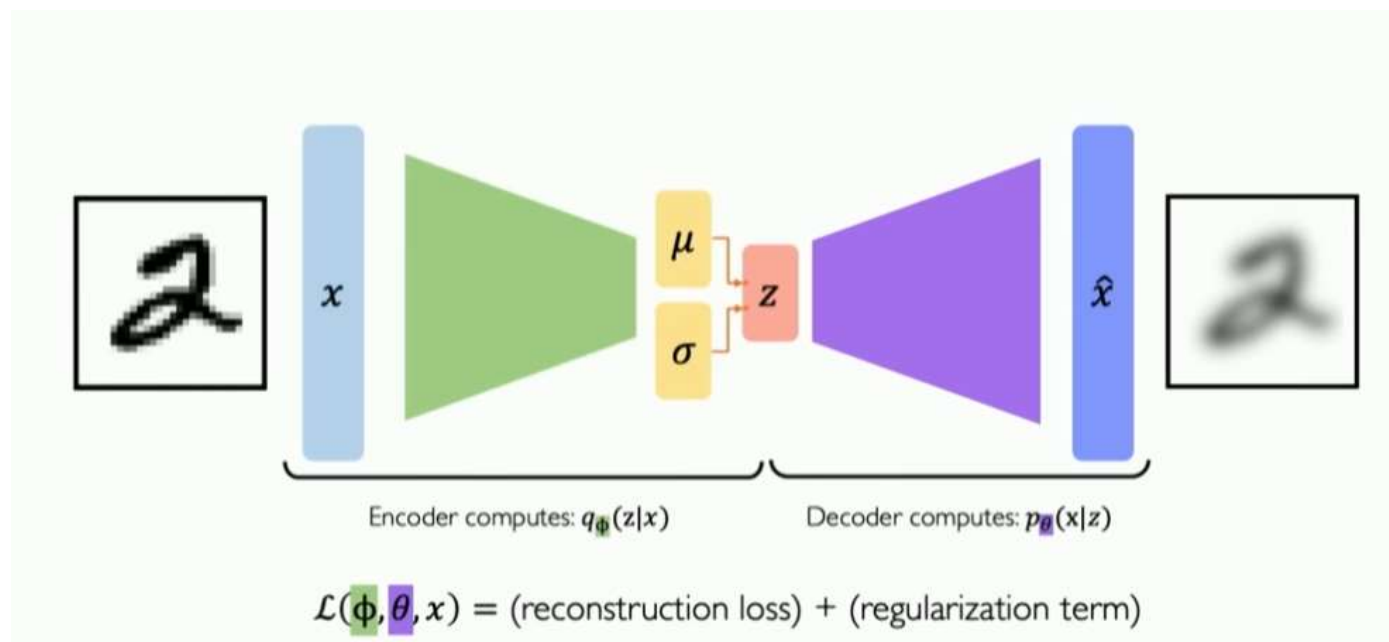
VAE OPTIMIZATION



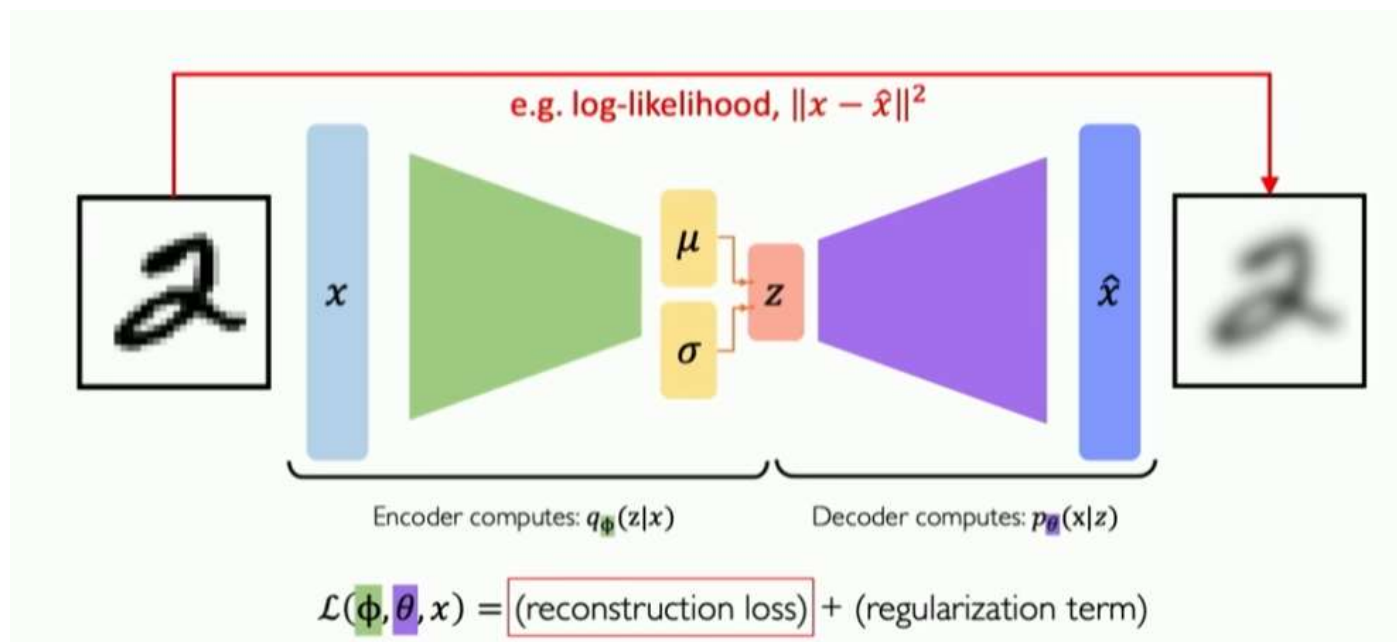
VAE OPTIMIZATION



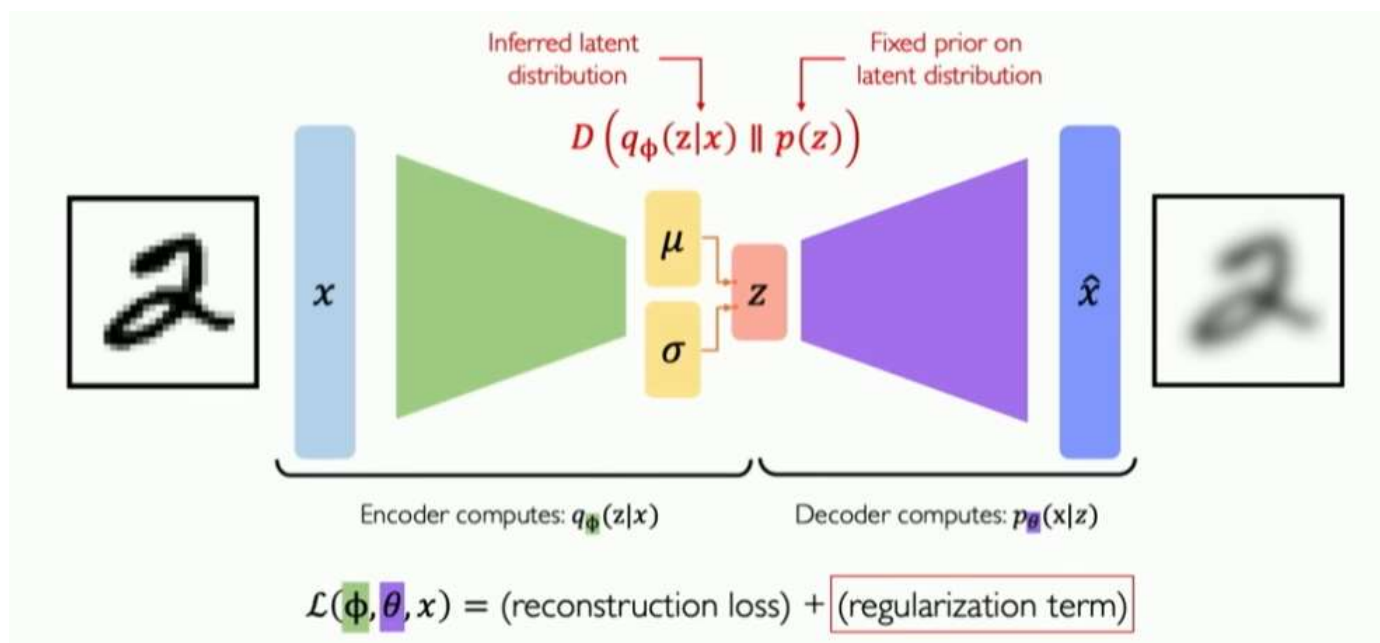
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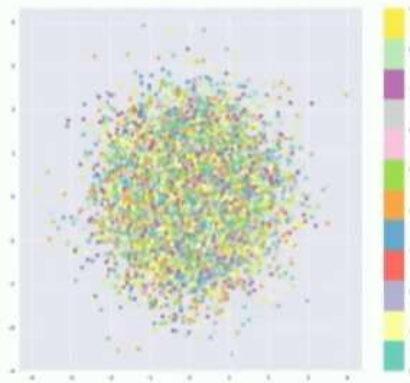


PRIORS ON THE LATENT DISTRIBUTION

$$D(q_{\phi}(z|x) \parallel p(z))$$

Inferred latent
distribution

Fixed prior on
latent distribution

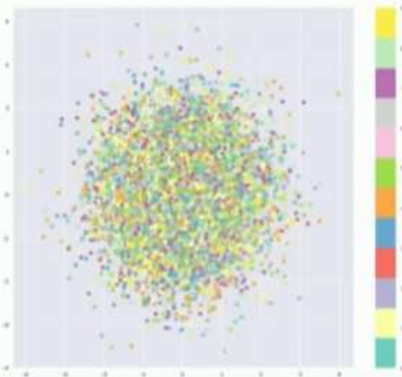


Common choice of prior – Normal Gaussian:

$$p(z) = \mathcal{N}(\mu = 0, \sigma^2 = 1)$$

- Encourages encodings to distribute evenly around the center of the latent space
- Penalize the network when it tries to "cheat" by clustering points in specific regions (i.e., by memorizing the data)

PRIORS ON THE LATENT DISTRIBUTION



$$D \left(q_{\phi}(z|x) \parallel p(z) \right)$$
$$= -\frac{1}{2} \sum_{j=0}^{k-1} (\sigma_j + \mu_j^2 - 1 - \log \sigma_j)$$

KL-divergence between the two distributions

Common choice of prior – Normal Gaussian:

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INTUITION ON REGULARIZATION AND THE NORMAL PRIOR

What properties do we want to achieve from regularization? 🤔

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1. **Continuity:** points that are close in latent space → similar content after decoding

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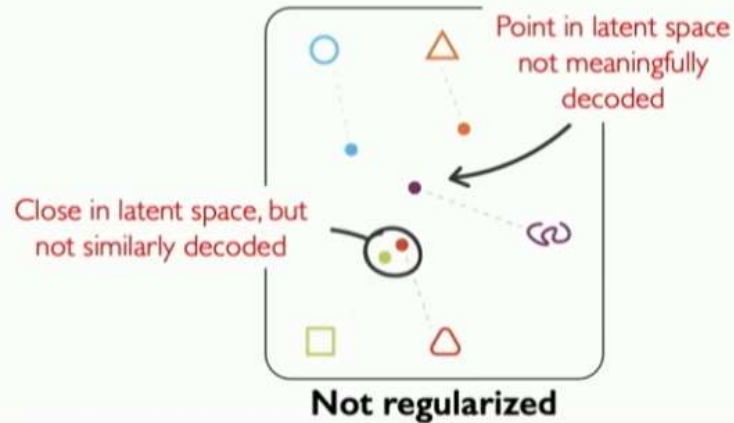
What properties do we want to achieve from regularization? 🤔

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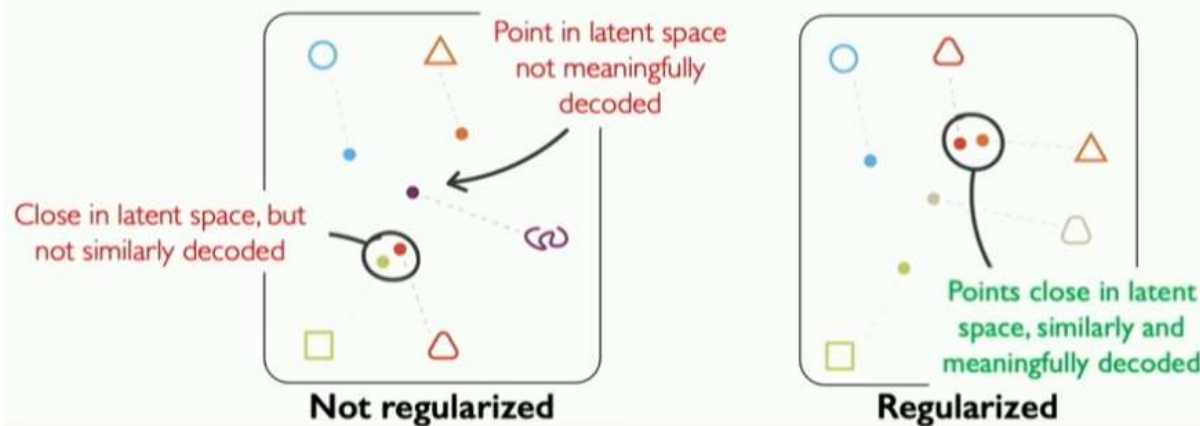
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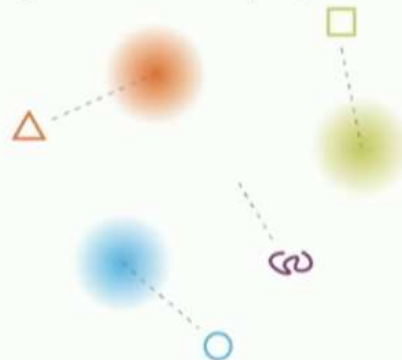
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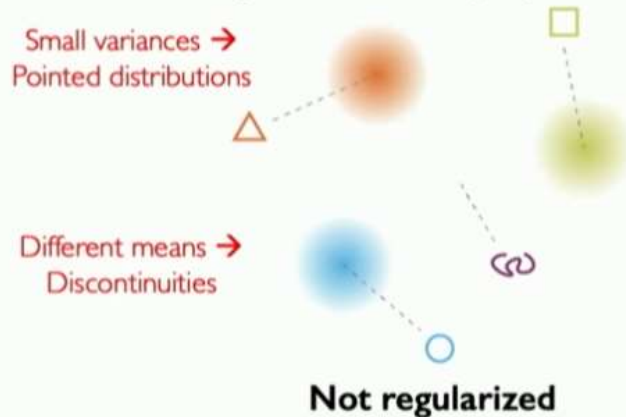


Not regularized

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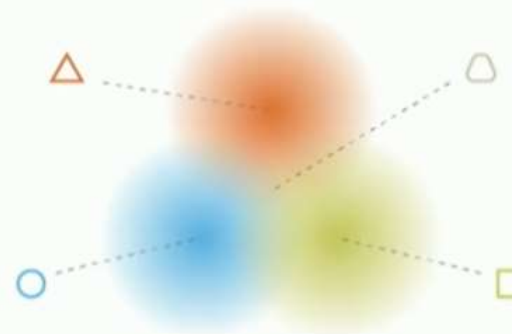
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Small variances $\rightarrow$
Pointed distributions

Different means $\rightarrow$
Discontinuities

Not regularized

Normal prior $\rightarrow$
continuity + completeness



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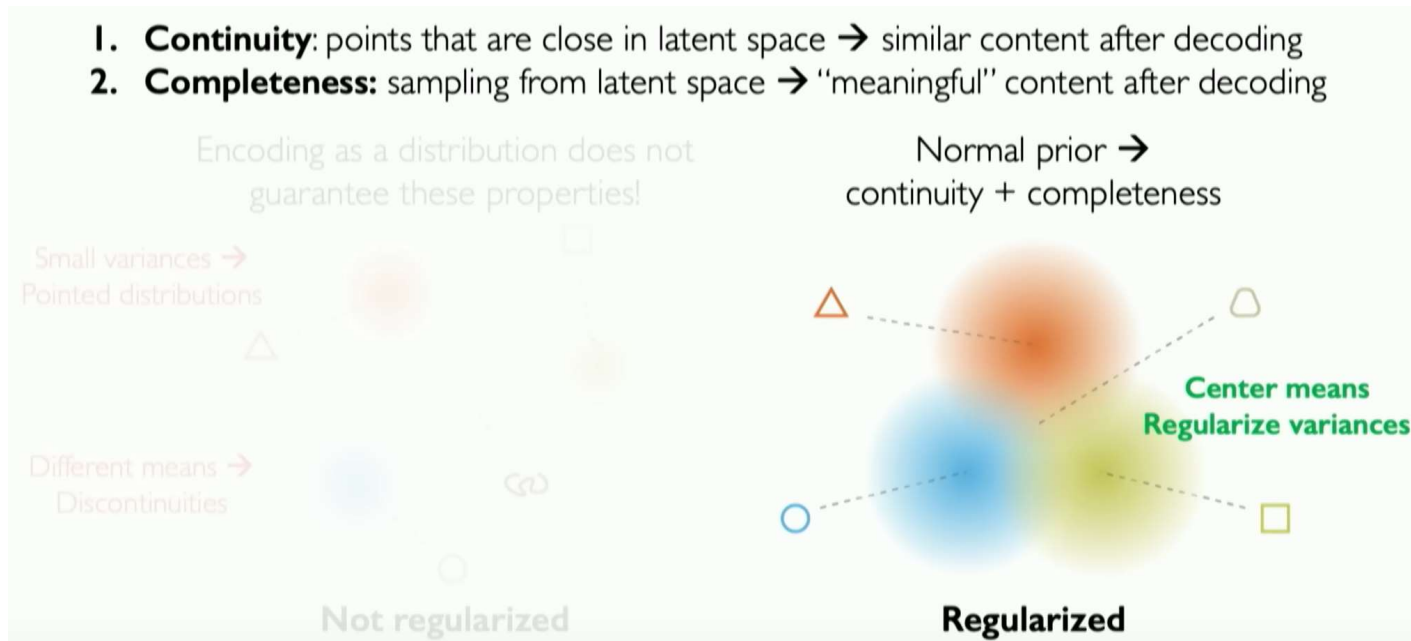
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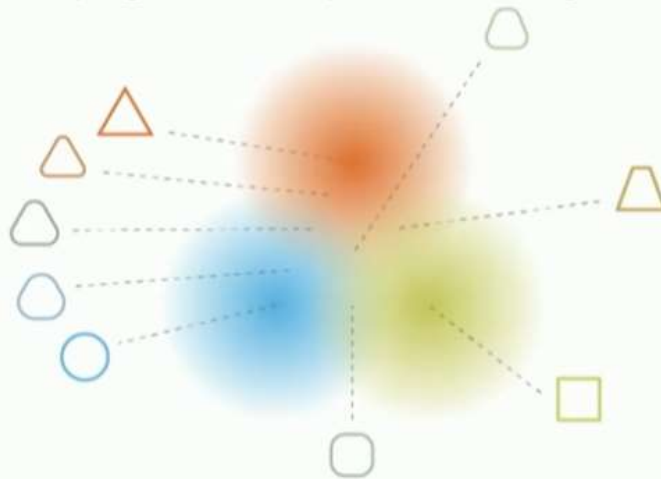
Center means
Regularize variances

Regularized



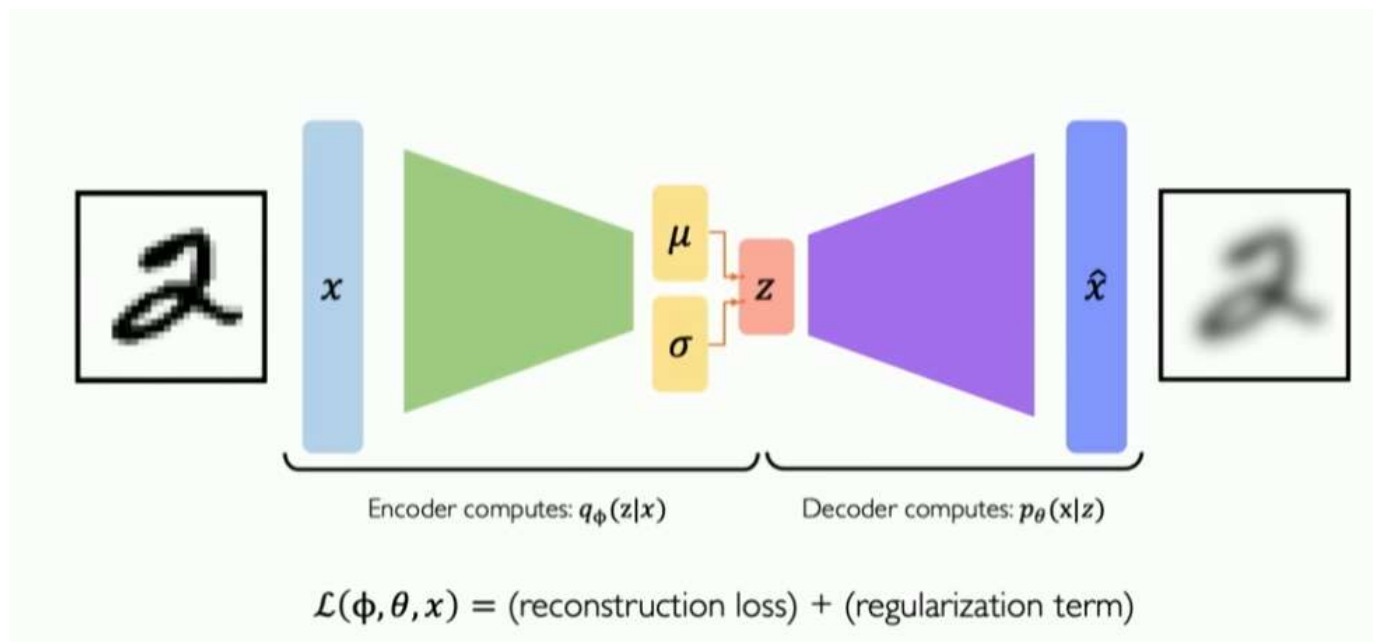
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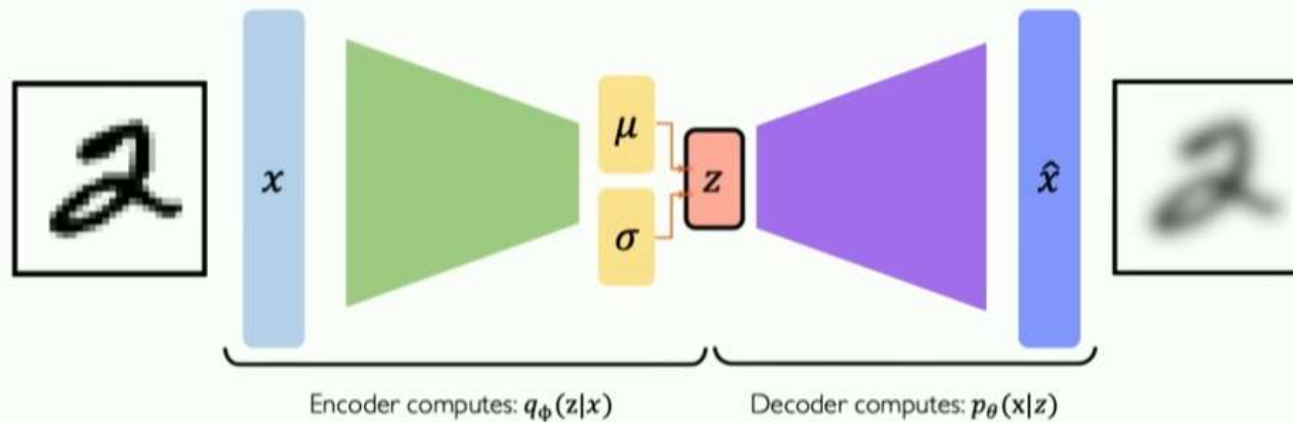
Regularization with Normal prior helps enforce **information gradient** in the latent space.

VAE COMPUTATION GRAPH



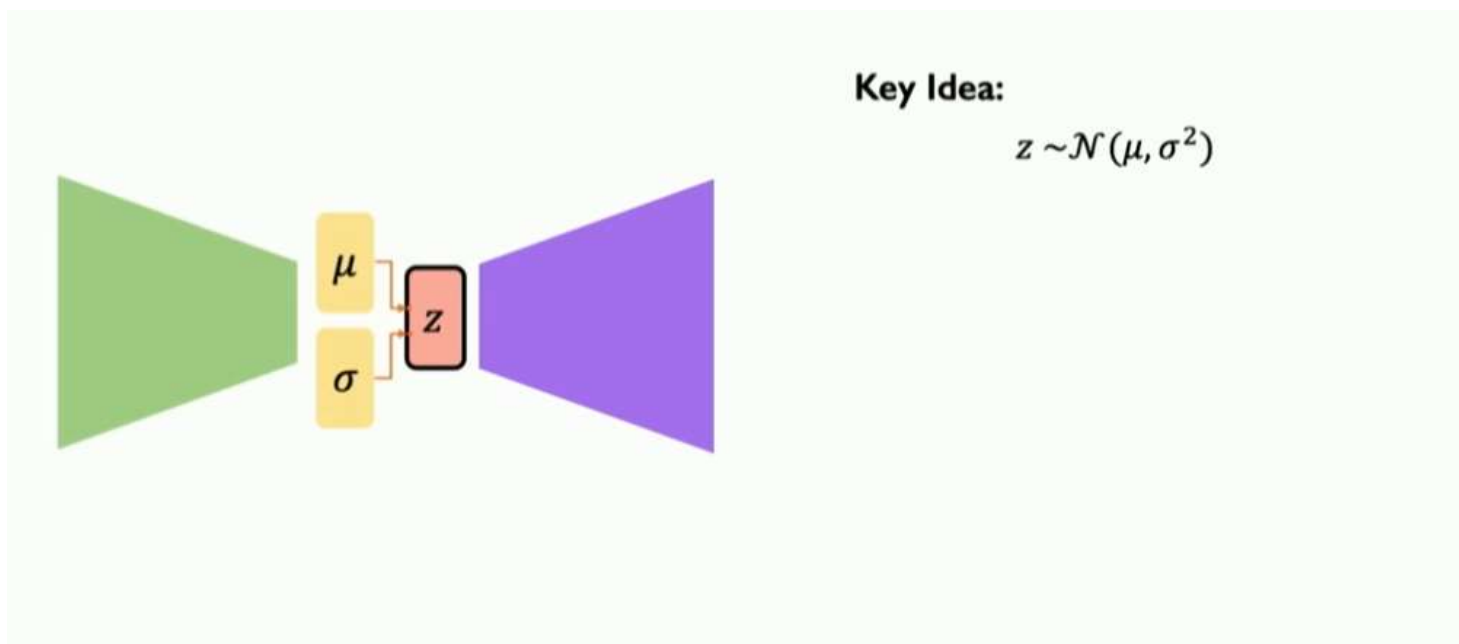
VAE COMPUTATION GRAPGH

Problem: We cannot backpropagate gradients through sampling layers!

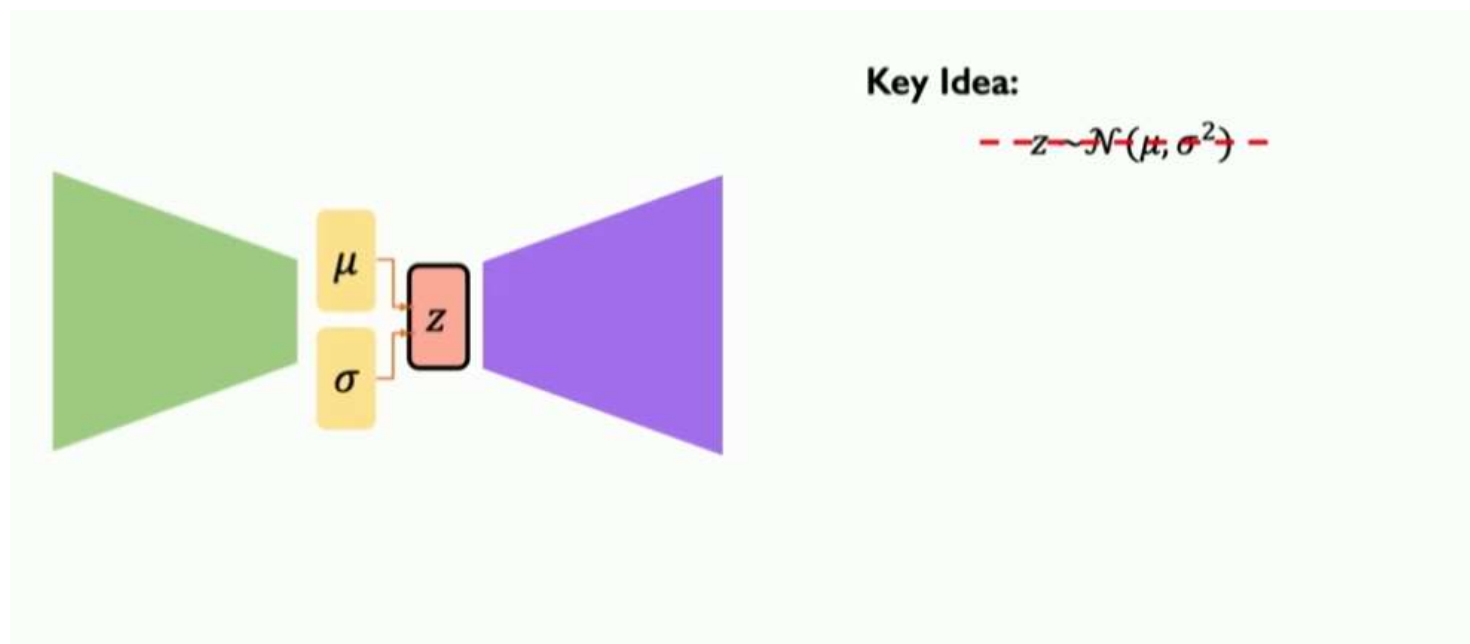


$$\mathcal{L}(\phi, \theta, x) = (\text{reconstruction loss}) + (\text{regularization term})$$

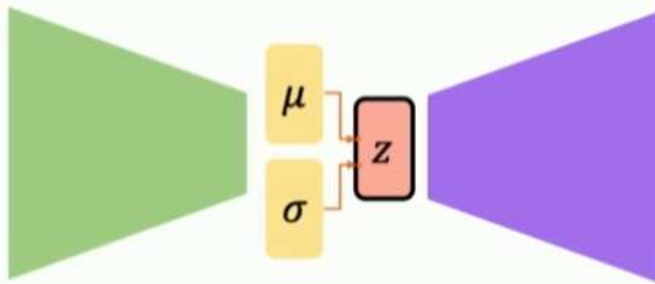
REPARAMETRIZING THE SAMPLING LAYER



REPARAMETRIZING THE SAMPLING LAYER



REPARAMETRIZING THE SAMPLING LAYER



Key Idea:

$$- - z \sim \mathcal{N}(\mu, \sigma^2) - -$$

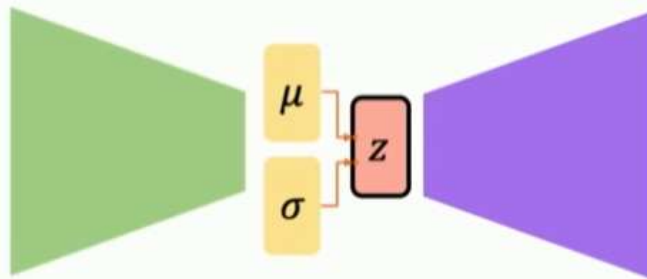
Consider the sampled latent vector z as a sum of

- a fixed μ vector,
- and fixed σ vector, scaled by random constants drawn from the prior distribution

$$\Rightarrow z = \mu + \sigma \odot \epsilon$$

where $\epsilon \sim \mathcal{N}(0,1)$

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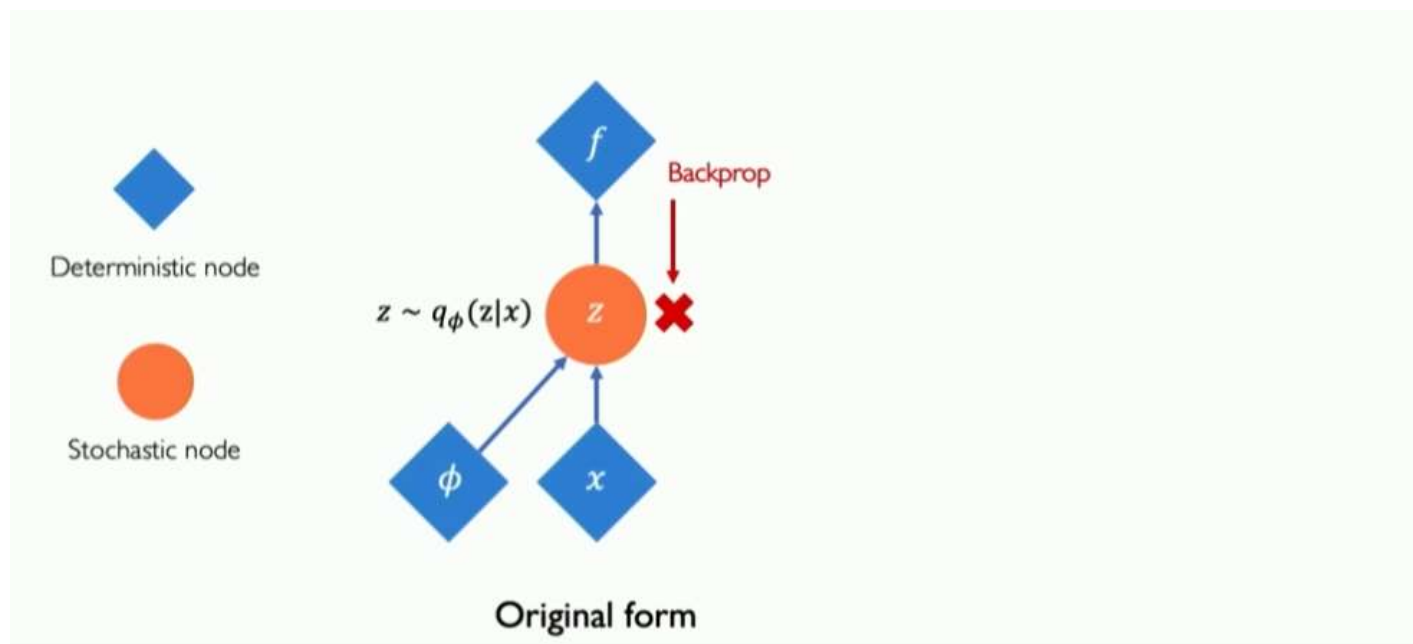
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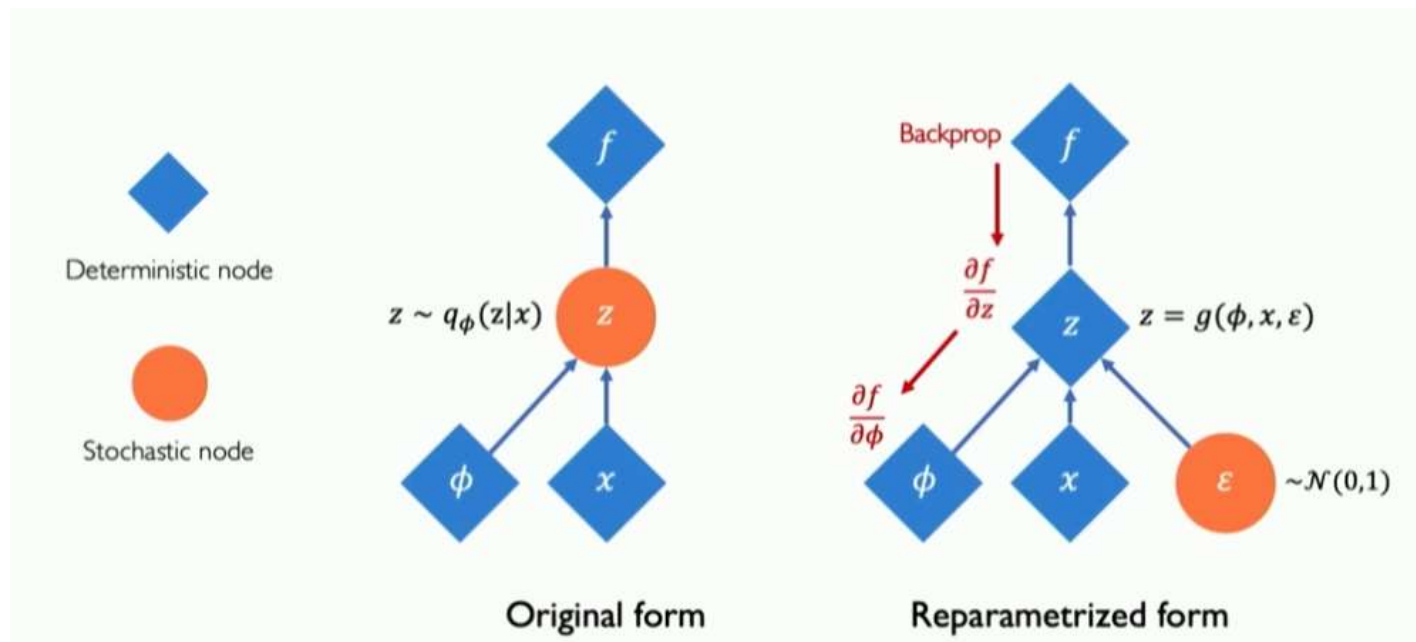
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MERCI